Demonstrating the physics of metamaterials with acoustic waves and soda cans

(Progress Report)

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Objective

1. To study physics of a soda can as a Helmholtz resonator by measuring the amplitude and phase response of the soda-can resonator as a function of the driving frequency.
2. To investigate the resonance frequency of the soda can for the different internal volumes.
3. To construct and characterize an acoustic antenna capable of redirecting sound preferentially in one direction.

Summary of Work Carried Out

Week 4

- First, corrected the python code and evaluated the obtained data via the PicoScope.
- As the data was not coming as expected, different ways were tried: changed distance of source and can, speaker orientation and amplitude of input signal.
- After this,we checked input signal of PicoScope with code and then we found out that the signal generated by PicoScope was a bit noisy.

Week 5

- To improve input signal quality and fine observation, small speaker, function generator and Oscilloscope were used.
- Recorded sharp resonance amplitudes; however, there was a systematic error in the phase measurements, crossing at around 110°, rather than theoretically crossing at 90°.
- Actually, noise was coming due to the vibrations and sound of oscilloscope. To resolve the issue, absorbent sheets were used.

Week 6

- To stablize the output signal and to get correct phase plot, high amplitude signals were used which surpressed the noise relatively.
- Finally, Data was obtained as expected with slight variations (at resonance peak phase came 90°).

- In past weeks, issues resolved regarding the stability of the output signal and its phase shift.

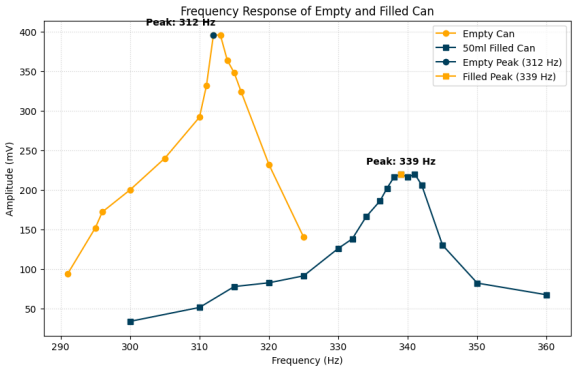


Figure 1: Frequency response curves showing peak resonance at 312 Hz for the empty can and 339 Hz for the 50 mL filled can.

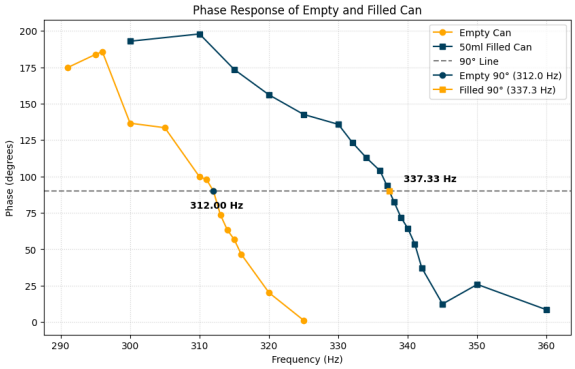


Figure 2: Phase response curves showing 90° phase crossings at 312.00 Hz (empty can) and 337.33 Hz (50 mL filled can).

Plan to complete during midsem break

- We would try observe the effect on resonance phenomena by varying the size of hole on the can's neck and would also try understand the behaviour of metamaterial by using multiple cans arranged in an array. We will construct and study an acoustic antenna using array of cans.
- Also, we will try to investigate the dependence of the resonance frequency on the opening area of the can by varying its opening size.